



COURSE SYLLABUS						
Cours e Code	COMP 015	Course Title	FUNDAMENTALS OF RESEARCH		Course Credit	3
Course Description		This course introduces students to the fundamental principles, processes, and ethics of research. It equips learners with the knowledge and skills to formulate research problems, design methodologies, and analyze data. Emphasis is placed on developing a research proposal that reflects academic rigor and ethical responsibility.				
Pre-Requisites			Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an Inclusive, Equitable, and Globally Relevant Polytechnic Education Towards National Development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

As a polytechnic state university, PUP shall develop its students to possess:

1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: - Offer curricula which are relevant and responsive to the changing needs of the industry and society; - Provide trainings to students that will enable them to become potent instruments to socio-economic development. - Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. - Provide state-of-the-arts facilities of learning to optimize student development; - Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; - Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; - Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; - Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; - Facilitate presentation of research outputs in international fora, and their publication in recognized journals; - Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, - Produce technologies for commercialization of livelihood improvement.



PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	Apply knowledge of computing, science, and mathematics appropriate to the discipline.									
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution									
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.									
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.									



	Integrate IT-based solutions into the user environment effectively.													
	Assist in the creation of an effective IT project plan.													
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession													
	Understand best practices and standards and their applications.													
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.													
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.													
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.													
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.													
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development													



	Alignment to PLOs							
	1	2	3	4	5	6	7	8
Explain the fundamental concepts, nature, and significance of research in the field of Information Technology and its contribution to innovation and problem-solving.								
Differentiate the various types, methods, and approaches of research applicable to computing and information systems.								
Formulate clear and researchable problems, questions, objectives, and hypotheses aligned with IT-related issues.								
Conduct a comprehensive and ethical review of related literature using credible print and digital sources, databases, and citation tools.								
Design appropriate research methodologies, data-gathering instruments, and analytical frameworks suited for IT-based research.								
Apply statistical and digital tools for organizing, interpreting, and presenting research data effectively.								
Demonstrate professional and ethical standards in the collection, analysis, and reporting of data, ensuring integrity and respect for intellectual property.								

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	Prepare and present a complete research proposal addressing an identified IT problem, following proper academic formatting and documentation standards.									
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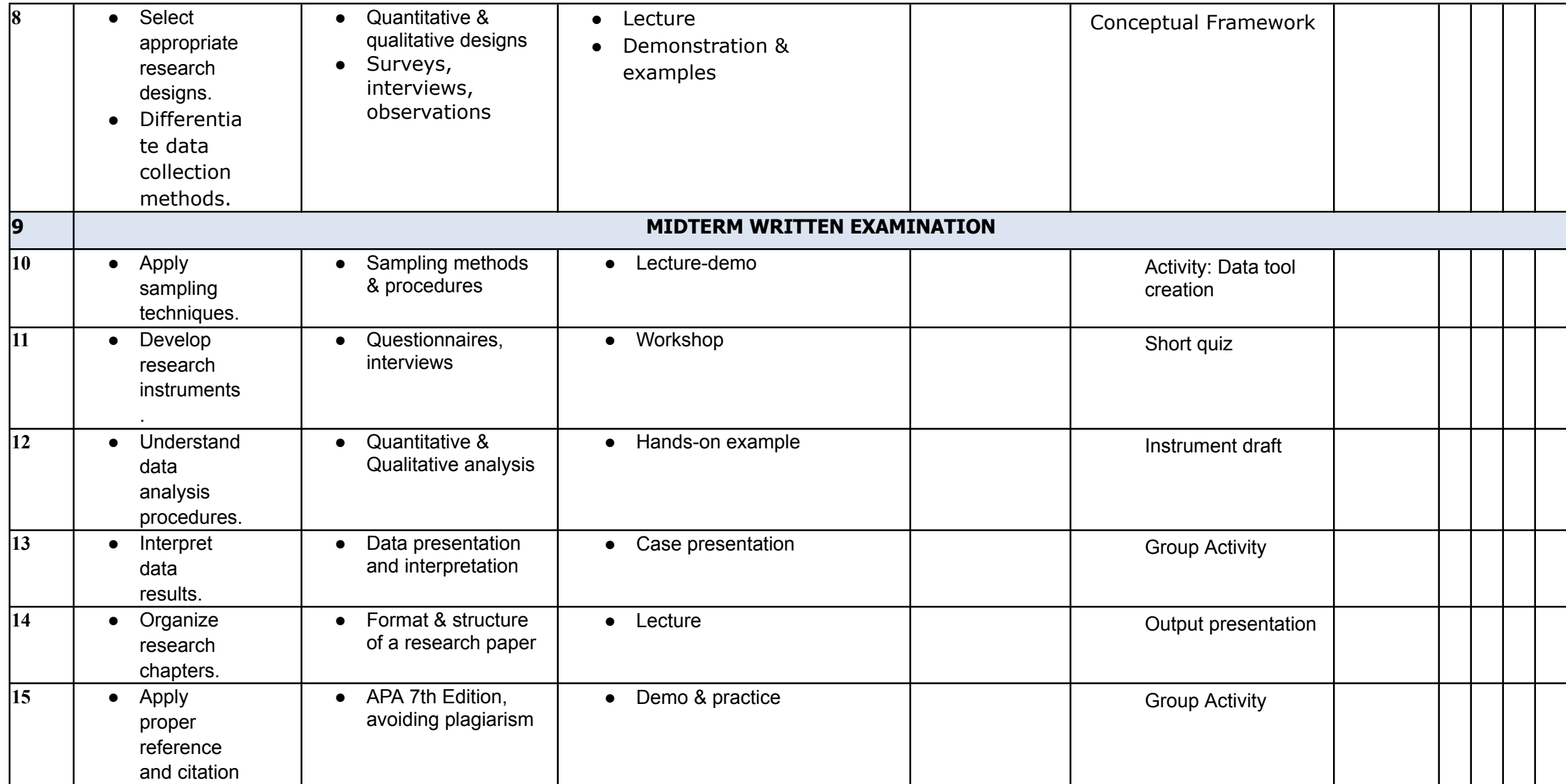


OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face, and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	Identify different types and classifications of research.	Basic vs. Applied, Qualitative vs. Quantitative	Group discussion		Short quiz					
3	Describe the research process.	Steps in Research Process	Interactive lecture		Quiz, participation					



4	Formulate a research problem and question.	Identifying and stating research problems	Workshop on problem statements		Reflection paper					
5	Write clear and measurable research objectives.	Formulating objectives and hypotheses	Lecture-workshop		Submission of research problem					
6	Conduct a brief review of related literature.	Sources, citation, paraphrasing	Hands-on library research		Research objective draft					
7	Identify and define variables and concepts.	Variables, indicators, conceptual framework	Lecture & sample analysis		RRL draft					



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	styles.									
16	• Write and revise the research manuscript.	• Drafting and revising papers	• Peer editing		Draft submission					
17	• Prepare and present a research proposal. • Reflect on research learning and ethics.	• Presentation skills, proposal defense • Research ethics, challenges, lessons learned	• Mock defense • Reflection session		Oral defense Final reflection paper					
18	FINAL WRITTEN EXAMINATION (PRACTICAL EXAMINATION)									



		Resources and Applying Strategies in Teaching Prose Competency 4: Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies ● Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		● Quizzes					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: Bouchrika, I. (2025). <i>How to write a research paper: Outline, format, publication types, ethics, and use of digital tools</i> . Research.com. https://research.com/research/how-to-write-a-research-paper Newhart, M., & Patten, M. L. (2023). <i>Understanding Research Methods: An Overview of the Essentials</i> (11th ed.). Routledge. Open Library Thomas, C. G. (2021). <i>Research Methodology and Scientific Writing</i> . Springer Cham. https://doi.org/10.1007/978-3-030-64865-7 SpringerLink Dermody, K., Farnum, C., Jakubek, D., Petropoulos, J.-A., Schmidt, J., & Steinberg, R. (2022). <i>Advanced Research Skills: Conducting Literature and Systematic Reviews</i> (2nd ed.) [Open Textbook]. Toronto Metropolitan University Library. pressbooks.library.torontomu.ca Pajo, B. (2022). <i>Introduction to Research Methods: A Hands-On Approach</i> (2nd ed.). SAGE Publications. collegepublishing.sagepub.com Mbanaso, U. M., Abrahams, L., & Okafor, K. C. (2023). <i>Research Techniques for Computer Science, Information Systems and Cybersecurity</i> . Springer Cham. https://doi.org/10.1007/978-3-031-30031-8 SpringerLink										



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students' Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



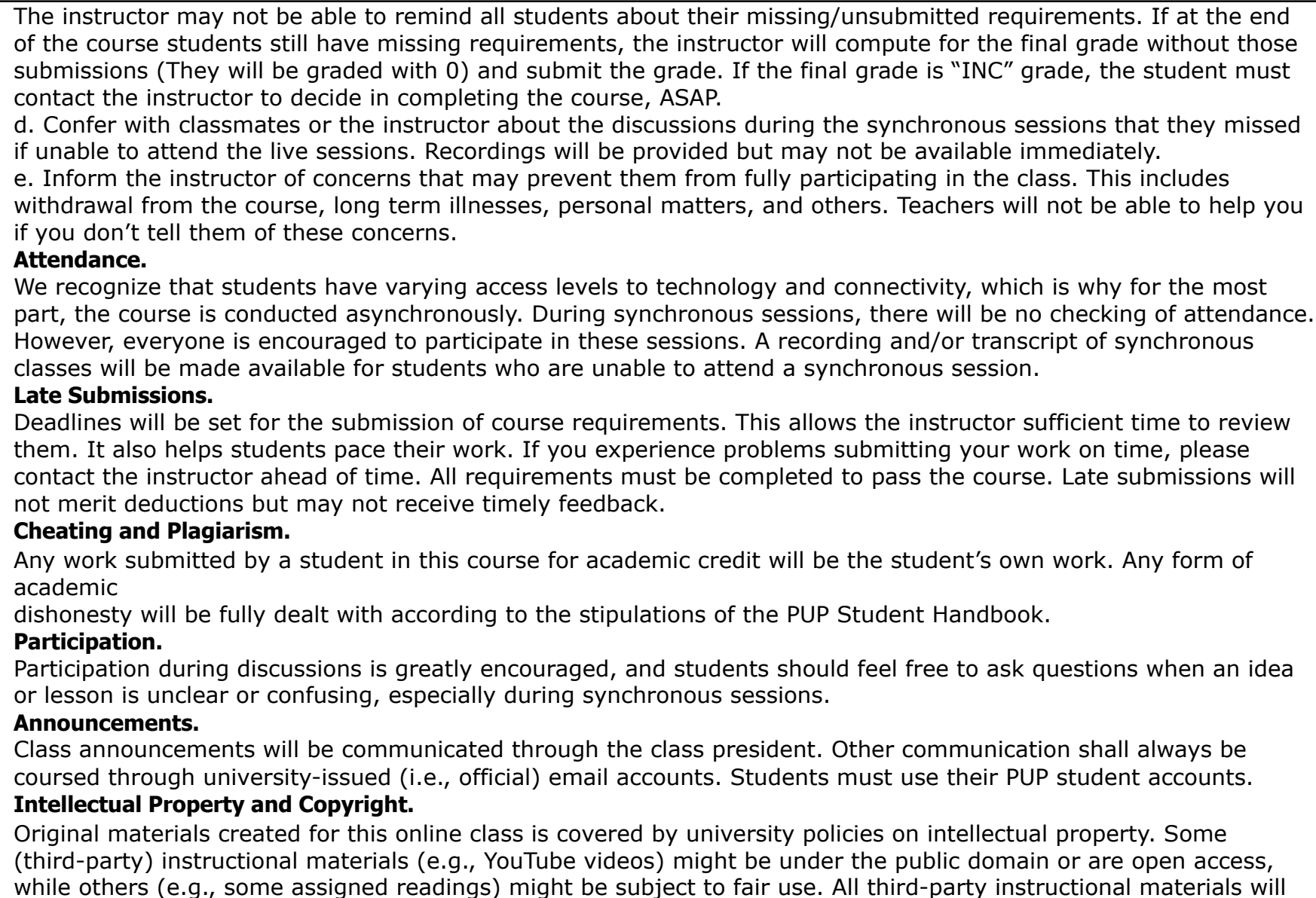
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	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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WORLD
UNIVERSITY
RANKINGS

2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



The WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI

Rank 39
STUDENT SUPPORT

Rank 50
SYMBOL/



STARS

★★★★★

★★★★★

★★★★★

Employability

Teaching

Social Responsibility

Online Learning

Academic Development

Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Internationalization



Times Higher Education
Impact Rankings 2024

DEBUTS AT 1501+



	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
Students are required to participate in all class discussions and complete assigned tasks, including quizzes, reflection papers, and written outputs. They must formulate a research problem, conduct a review of related literature, and design appropriate research instruments. The course culminates in the preparation and oral presentation of a complete research proposal demonstrating academic rigor and ethical standards.	Mid Term	Final Term
	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper
	Midterm Examinations 30%	Lesson Plan 30%
	100%	Final Examinations (Teaching Demonstration) 100%
Final Grade = (Midterm Grade + Final Term Grade) / 2		
CLASS INFORMATION		FACULTY INFORMATION
Course, Year and Section: BSIT 3 – 1		Name of Faculty: ALMIRANEZ, GECILIE C.
Day and Time:		Consultation Time:
Room:		Office Tel. No./ Mobile Phone No.
School Year:	Semester:	E-Mail Address:



Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	DR. RYHAN V. MOLINAR Campus Director	DR. ARMANDO A. TORRES Vice President for Campuses